

PRACTICAL WEEK-3

TASK1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Parent process started.\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("\n--- Child Process ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());
        printf("Child State : Running\n");

        sleep(5);

        printf("Child State : Terminated\n");
        exit(0);
    }
    else
    {
        printf("\n--- Parent Process ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        printf("Parent State: Running\n");
    }
}
```

```

        sleep(5);

        printf("Child State : Terminated\n");
        exit(0);
    }
    else
    {
        printf("\n--- Parent Process ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        printf("Parent State: Running\n");

        wait(NULL);

        printf("Parent State: Child terminated, parent continuing.\n");
    }

    return 0;
}

```

```

laasya@laasya:~/week3$ gcc task3.c -o task3
laasya@laasya:~/week3$ ./task3
Parent process started.
Parent PID : 13241
Parent PPID : 13199

--- Parent Process ---
Parent PID : 13241
Child PID : 13242
Parent State: Running

--- Child Process ---
Child PID : 13242
Child PPID : 13241
Child State : Running
Child State : Terminated
Parent State: Child terminated, parent continuing.
laasya@laasya:~/week3$ |

```

TASK2: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
laasya@laasya:~/week3$ cat task4.c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define MAX_HISTORY 10
#define MAX_LENGTH 100

char history[MAX_HISTORY][MAX_LENGTH];
int history_count = 0;

void clear_line()
{
    printf("\r\033[K");
    printf("myshell> ");
}

int main()
{
    struct termios oldt, newt;
    char buffer[MAX_LENGTH];
    char ch;
    int index;
    int history_position;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;

    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1)
    {
        index = 0;
        history_position = history_count;

        memset(buffer, 0, sizeof(buffer));

        printf("myshell> ");
```

```
index = 0;
history_position = history_count;

memset(buffer, 0, sizeof(buffer));

printf("myshell> ");
fflush(stdout);

while (1)
{
    ch = getchar();

    /* Enter key */
    if (ch == '\n')
    {
        buffer[index] = '\0';

        printf("\n");

        if (strlen(buffer) > 0)
        {
            if (strcmp(buffer, "exit") == 0)
            {
                tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
                printf("Exiting...\n");
                return 0;
            }

            if (history_count < MAX_HISTORY)
            {
                strcpy(history[history_count], buffer);
                history_count++;
            }
            else
            {
                for (int i = 1; i < MAX_HISTORY; i++)
                {
                    strcpy(history[i - 1], history[i]);
                }
            }
        }
    }
}
```

```

        printf("You entered: %s\n", buffer);
        break;
    }

    /* Backspace */
    else if (ch == 127 || ch == 8)
    {
        if (index > 0)
        {
            index--;
            buffer[index] = '\0';

            printf("\b \b");
            fflush(stdout);
        }
    }

    /* Escape sequence */
    else if (ch == 27)
    {
        char ch1 = getchar();

        if (ch1 == '[')
        {
            char ch2 = getchar();

            /* Up arrow */
            if (ch2 == 'A')
            {
                if (history_count > 0 && history_position > 0)
                {
                    history_position--;

                    strcpy(buffer, history[history_position]);
                    index = strlen(buffer);

                    clear_line();
                    printf("%s", buffer);
                }
            }
        }
    }
}

```

```

        fflush(stdout);
    }
}

/* Down arrow */
else if (ch2 == 'B')
{
    if (history_position < history_count - 1)
    {
        history_position++;

        strcpy(buffer, history[history_position]);
        index = strlen(buffer);

        clear_line();
        printf("%s", buffer);
        fflush(stdout);
    }
    else
    {
        history_position = history_count;
        index = 0;
        buffer[0] = '\0';

        clear_line();
        fflush(stdout);
    }
}

}

/* Normal character */
else if (index < MAX_LENGTH - 1)
{
    buffer[index] = ch;
    index++;

    putchar(ch);
    fflush(stdout);
}

```

```

        clear_line();
        fflush(stdout);
    }
}

/* Normal character */
else if (index < MAX_LENGTH - 1)
{
    buffer[index] = ch;
    index++;

    putchar(ch);
    fflush(stdout);
}
}

tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

return 0;
}
laasya@laasya:~/week3$ |

```

```

laasya@laasya:~/week3$ gcc task4.c -o task4
laasya@laasya:~/week3$ ./task4
myshell> ls
You entered: ls
myshell> pwd
You entered: pwd
myshell> exit
Exiting...
laasya@laasya:~/week3$ sleep 30 &
[1] 13279
laasya@laasya:~/week3$ ps -o pid,ppid,state,cmd -p 13279
  PID   PPID  S  CMD
  13279   13199 S  sleep 30
laasya@laasya:~/week3$ cat /proc/13279/status
cat: /proc/13279/status: No such file or directory
[1]+  Done                  sleep 30
laasya@laasya:~/week3$ top
top - 17:12:49 up 29 min,  1 user,  load average: 0.01, 0.01, 0.02
Tasks:  27 total,   1 running,  26 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.1 sy,   0.0 ni, 99.9 id,   0.0 wa,   0.0 hi,   0.0 si,   0.
MiB Mem :   7762.6 total,   7122.2 free,    549.8 used,    262.2 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,     0.0 used.   7212.9 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
6622	root	20	0	25124	5552	4284	S	0.3	0.1	0:00.41
1	root	20	0	22128	10748	6944	S	0.0	0.1	0:04.40
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.00
7	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00
177	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.02
178	message+	20	0	9828	5628	4640	S	0.0	0.1	0:00.98
185	root	20	0	17968	8860	7812	S	0.0	0.1	0:00.33
205	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00
224	root	20	0	107020	23172	13708	S	0.0	0.3	0:00.14
299	root	20	0	3188	1104	980	S	0.0	0.0	0:00.00
300	root	20	0	3204	1244	1108	S	0.0	0.0	0:00.21
301	laasya	20	0	6080	5416	3700	S	0.0	0.1	0:00.12
302	root	20	0	6700	4656	3876	S	0.0	0.1	0:00.01
345	laasya	20	0	20128	7904	6088	S	0.0	0.1	0:00.17
346	laasya	20	0	21148	3592	1872	S	0.0	0.0	0:00.00
370	laasya	20	0	6080	5360	3696	S	0.0	0.1	0:00.02
2310	root	19	-1	50376	14060	12964	S	0.0	0.2	0:00.23
6105	syslog	20	0	222516	5100	4168	S	0.0	0.1	0:00.05
6569	systemd+	20	0	91036	6012	5044	S	0.0	0.1	0:00.11
6898	systemd+	20	0	21468	9220	6960	S	0.0	0.1	0:00.12
10909	polkitd	20	0	308172	6700	5800	S	0.0	0.1	0:00.11
13191	root	20	0	3188	1112	980	S	0.0	0.0	0:00.00
13193	root	20	0	3204	1252	1108	S	0.0	0.0	0:00.05
13199	laasya	20	0	6596	5896	3652	S	0.0	0.1	0:00.08
13288	laasya	20	0	9300	5736	3552	R	0.0	0.1	0:00.01
13289	root	20	0	25128	3540	2240	S	0.0	0.0	0:00.00
13290	root	20	0	25128	3540	2240	S	0.0	0.0	0:00.00

```

laasya@laasya:~/week3$ ps -p 13279
  PID TTY          TIME CMD
laasya@laasya:~/week3$ |

```